

```
gireesh@Bhashwanth:~$ make --version
GNU Make 4.4.1
Built for x86_64-pc-linux-gnu
Copyright (C) 1988-2023 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.htm
l>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
gireesh@Bhashwanth:~$ git --version
git version 2.53.0
gireesh@Bhashwanth:~$ cd ~
gireesh@Bhashwanth:~$ mkdir OSLab
mkdir: OSLab: File exists
gireesh@Bhashwanth:~$ cd OSLab
gireesh@Bhashwanth:~/OSLab$ mkdir src include bin obj
gireesh@Bhashwanth:~/OSLab$ ls
bin      copy.txt  hello    obj      process.c  states
copy     filecopy  hello.c  output.txt  sample.txt  states.c
copy.c   filecopy.c  include  process    src
gireesh@Bhashwanth:~/OSLab$ git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warni
ng,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command
:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName fal
se"
Initialized empty Git repository in /home/gireesh/OSLab/.git/
gireesh@Bhashwanth:~/OSLab$ git status
On branch master
```

```
nothing added to commit but untracked files present (use "git add" to track)
gireesh@Bhashwanth:~/OSLab$ git config --global user.name "Gireesh"
gireesh@Bhashwanth:~/OSLab$ git config --global user.email "your@email.com"
gireesh@Bhashwanth:~/OSLab$ git config --list
user.name=Gireesh
user.email=your@email.com
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
gireesh@Bhashwanth:~/OSLab$ git config --global user.email "chowdarygireesh8@gmail.com"
gireesh@Bhashwanth:~/OSLab$ git config --list
user.name=Gireesh
user.email=chowdarygireesh8@gmail.com
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
gireesh@Bhashwanth:~/OSLab$ nano src/main.c
gireesh@Bhashwanth:~/OSLab$ gcc src/main.c -o bin/main
gireesh@Bhashwanth:~/OSLab$ ./bin/main
Hello, Operating Systems Lab!
gireesh@Bhashwanth:~/OSLab$ nano src/fork_demo.c
gireesh@Bhashwanth:~/OSLab$ gcc src/fork_demo.c -o bin/fork_demo
gireesh@Bhashwanth:~/OSLab$ ./bin/fork_demo
Before fork()
Parent process
Parent PID = 3573
Child PID = 3574
Child process
Child PID = 3574
Parent PID = 3573
```

gireesh@Bhashwanth: ~/OSL: ×

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GNU nano 8.7.1

src/fork_demo.c *

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>

int main() {
    pid_t pid;

    printf("Before fork()\n");

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        printf("Child process\n");
        printf("Child PID = %d\n", getpid());
        printf("Parent PID = %d\n", getppid());
    } else {
        printf("Parent process\n");
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
    }

    return 0;
}
```



```
gireesh@Bhashwanth: ~/OSLab × + ▾
GNU nano 8.7.1 src/multiple_fork.c *
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("Original process PID = %d\n", getpid());

    fork();
    fork();

    printf("Process PID = %d, Parent PID = %d\n",
           getpid(), getppid());

    return 0;
}
```

```
gireesh@Bhashwanth:~/OSLab$ pid = fork();
-bash: syntax error near unexpected token `('
gireesh@Bhashwanth:~/OSLab$ nano src/exec_demo.c
gireesh@Bhashwanth:~/OSLab$ gcc src/exec_demo.c -o bin/exec_demo
gireesh@Bhashwanth:~/OSLab$ ./bin/exec_demo
Before exec()
total 128
-rw-r--r-- 1 gireesh gireesh      0 Aug 17 03:31 0
drwxr-xr-x 2 gireesh gireesh  4096 Aug 17 05:05 bin
-rwxr-xr-x 1 gireesh gireesh 16256 Aug 15 06:46 copy
-rw-r--r-- 1 gireesh gireesh  1093 Aug 15 06:45 copy.c
-rw-r--r-- 1 gireesh gireesh   160 Aug 14 17:53 copy.txt
-rwxr-xr-x 1 gireesh gireesh 16216 Aug 14 17:53 filecopy
-rw-r--r-- 1 gireesh gireesh  1247 Aug 14 17:51 filecopy.c
-rwxr-xr-x 1 gireesh gireesh 15952 Aug 14 06:39 hello
-rw-r--r-- 1 gireesh gireesh    79 Aug 14 06:38 hello.c
drwxr-xr-x 2 gireesh gireesh  4096 Aug 17 03:21 include
drwxr-xr-x 2 gireesh gireesh  4096 Aug 17 03:21 obj
-rw-r--r-- 1 gireesh gireesh    12 Aug 15 06:47 output.txt
-rwxr-xr-x 1 gireesh gireesh 16248 Aug 15 07:02 process
-rw-r--r-- 1 gireesh gireesh   885 Aug 15 07:01 process.c
-rw-r--r-- 1 gireesh gireesh    12 Aug 15 06:46 sample.txt
drwxr-xr-x 2 gireesh gireesh  4096 Aug 17 05:05 src
-rwxr-xr-x 1 gireesh gireesh 16080 Aug 15 07:15 states
-rw-r--r-- 1 gireesh gireesh   304 Aug 15 07:15 states.c
gireesh@Bhashwanth:~/OSLab$ nano src/parent_child.c
gireesh@Bhashwanth:~/OSLab$ gcc src/parent_child.c -o bin/parent_child
gireesh@Bhashwanth:~/OSLab$ ^C
gireesh@Bhashwanth:~/OSLab$ ./bin/parent_child
Parent process
Parent PID = 4146
Child PID = 4147
Child process
Child PID = 4147
Parent PID = 4146
Child process finished
gireesh@Bhashwanth:~/OSLab$ nano src/fork_exec.c
```

GNU nano 8.7.1

src/parent_child.c *

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    pid = fork();

    if (pid < 0) {
        printf("Fork failed\n");
        return 1;
    }

    if (pid == 0) {
        printf("Child process\n");
        printf("Child PID = %d\n", getpid());
        printf("Parent PID = %d\n", getppid());
    }
    else {
        printf("Parent process\n");
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);

        wait(NULL);

        printf("Child process finished\n");
    }

    return 0;
}
```

```

Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
gireesh@Bhashwanth:~/OSLab$ ps
  PID TTY          TIME CMD
   719 pts/2    00:00:00 bash
  4257 pts/2    00:00:00 ps
gireesh@Bhashwanth:~/OSLab$ ps -f
UID          PID    PPID  C STIME TTY          TIME CMD
gireesh       719      717  0 03:33 pts/2    00:00:00 -bash
gireesh      4267      719  0 05:08 pts/2    00:00:00 ps -f
gireesh@Bhashwanth:~/OSLab$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1         0  0 03:06 ?        00:00:02 /sbin/init
root           2         1  0 03:06 hvc0     00:00:00 /init
root           6         2  0 03:06 hvc0     00:00:00 plan9 --control-socket 7
root          46         1  0 03:06 ?        00:00:00 /usr/lib/systemd/systemd
systemd+      82         1  0 03:06 ?        00:00:00 /usr/lib/systemd/systemd
root          94         1  0 03:06 ?        00:00:02 /usr/lib/systemd/systemd
root         147         1  0 03:06 ?        00:00:00 /bin/sh /usr/lib/systemd
root         160         1  0 03:06 ?        00:00:00 /usr/sbin/cron -f -P
message+     161         1  0 03:06 ?        00:00:00 @dbus-daemon --system --
root         168         1  0 03:06 ?        00:00:00 /usr/bin/python3 /usr/bi
root         178         1  0 03:06 ?        00:00:00 /usr/lib/systemd/systemd
_chrony      265        147  0 03:06 ?        00:00:00 /usr/sbin/chronyd -n -F
_chrony      280        265  0 03:06 ?        00:00:00 /usr/sbin/chronyd -n -F
root         290         1  0 03:06 ?        00:00:00 /usr/bin/python3 /usr/sh
root         294         1  0 03:06 tty1     00:00:00 /usr/sbin/agetty --nores
root         330         2  0 03:06 ?        00:00:00 /init
root         334        330  0 03:06 ?        00:00:00 /init
gireesh      341        334  0 03:06 pts/0    00:00:00 -bash
root         342         2  0 03:06 ?        00:00:00 login -- gireesh
gireesh      385         1  0 03:06 ?        00:00:00 /usr/lib/systemd/systemd
gireesh      389        385  0 03:06 ?        00:00:00 (sd-pam)
gireesh      416        342  0 03:06 pts/1    00:00:00 -bash
root         715         2  0 03:33 ?        00:00:00 /init
root         717        715  0 03:33 ?        00:00:00 /init
gireesh      719        717  0 03:33 pts/2    00:00:00 -bash
polkitd       907         1  0 03:37 ?        00:00:00 /usr/lib/polkit-1/polkit
syslog       3150         1  0 03:38 ?        00:00:00 /usr/sbin/rsyslogd -n -i
gireesh      4279        719  0 05:08 pts/2    00:00:00 ps -ef
gireesh@Bhashwanth:~/OSLab$ pstree

```

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    pid = fork();

    if (pid < 0) {
        printf("Fork failed\n");
        return 1;
    }

    if (pid == 0) {
        printf("Child: executing ls...\n");

        execl("/bin/ls", "ls", "-l", NULL);

        perror("exec failed");
    }
    else {
        printf("Parent: waiting for child...\n");

        wait(NULL);

        printf("Parent: child finished\n");
    }

    return 0;
}
```



```

gireesh 4279 719 0 03:08 pts/2 00:00:00 ps =e+
gireesh@Bhashwanth:~/OSLab$ pstree
systemd├─agetty
│      └─chronyd-starter─chronyd─chronyd
│
│      └─cron
│
│      └─dbus-daemon
│
│      └─init-systemd(Ub)─SessionLeader─Relay(341)─bash
│                          └─SessionLeader─Relay(719)─bash─pstree
│                          └─init─{init}
│                          └─login─bash
│                          └─{init-systemd(Ub)}
│
│      └─networkd-dispat
│
│      └─polkitd─3*[{polkitd}]
│
│      └─rsyslogd─3*[{rsyslogd}]
│
│      └─systemd─(sd-pam)
│
│      └─systemd-journal
│
│      └─systemd-logind
│
│      └─systemd-resolve
│
│      └─systemd-udev
│
│      └─unattended-upgr─{unattended-upgr}

```

```

gireesh@Bhashwanth:~/OSLab$ pstree -p
systemd(1)─agetty(294)
│          └─chronyd-starter(147)─chronyd(265)─chronyd(280)
│
│          └─cron(160)
│
│          └─dbus-daemon(161)
│
│          └─init-systemd(Ub(2)─SessionLeader(330)─Relay(341)(334)─ba+
│                                └─SessionLeader(715)─Relay(719)(717)─ba+
│                                └─init(6)─{init}(7)
│                                └─login(342)─bash(416)
│                                └─{init-systemd(Ub)}(8)
│
│          └─networkd-dispat(168)
│
│          └─polkitd(907)─{polkitd}(908)
│                        └─{polkitd}(910)
│                        └─{polkitd}(911)
│
│          └─rsyslogd(3150)─{rsyslogd}(3151)
│                          └─{rsyslogd}(3152)
│                          └─{rsyslogd}(3153)
│
│          └─systemd(385)─(sd-pam)(389)
│
│          └─systemd-journal(46)
│
│          └─systemd-logind(178)
│
│          └─systemd-resolve(82)

```

